Parts of a whole

1 Complete the following statements:

a 30% is the same as ⬚ out of 100 and ⬚ out of 10.

b 15% is the same as ⬚ out of 100 and ⬚ out of 20.

c 75% is the same as ⬚ out of 100 and ⬚ out of 4.

d 80% is the same as ⬚ out of 100 and ⬚ out of 5.

e 50% is the same as ⬚ out of 100 and ⬚ out of 2.

f 62% is the same as ⬚ out of 100 and ⬚ out of 50.

2 Express the following percentages as a fraction:

i With a denominator of 100.

ii In simplest form.

a 40% **b** 65% **c** 30% **d** 16%

e 25% **f** 62.5% **g** 75% **h** 35%

3 Express the following quantities as a:

i Fraction

ii Percentage

a 21 out of 50 **b** 36 out of 45 **c** 4 out of 80 **d** 28 out of 56

e 42 out of 24 **f** 44 out of 110 **g** 64 out of 40 **h** 21 out of 175

Quantities as percentages

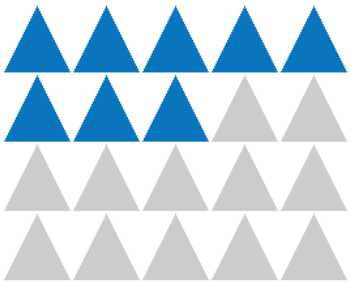
4 Calculate 15 as a percentage of 31. Round to two decimal places.

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5

a What percentage of the triangle are shaded?

i



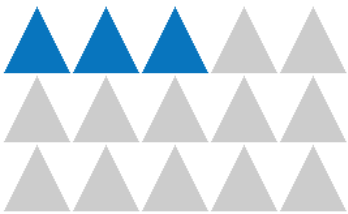
ii



iii

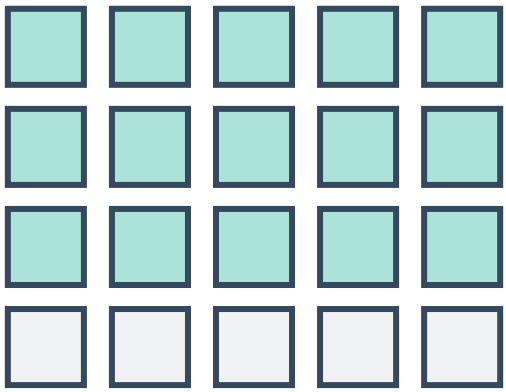


iv

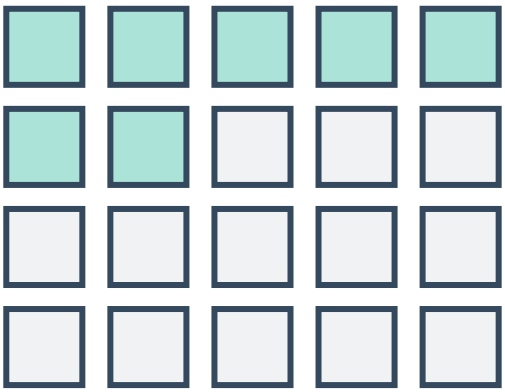


b What percentage of the squares are shaded?

i



ii



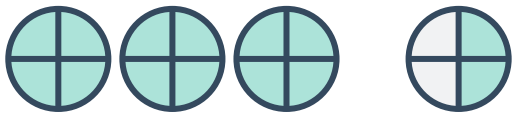
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c What percentage of one whole circle is shaded in each set?

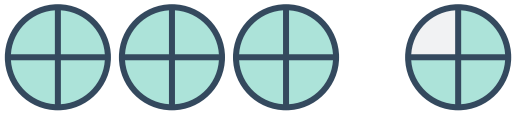
i



ii



iii



6 Write the following quantities as percentages:

a 36 out of 200 **b** 10 out of 16 **c** 9 out of 27 **d** 17 out of 68

7 Find:

a 27 as a percentage of 150.

b 81 as a percentage of 150.

c 34 as a percentage of 56.

d 12 as a percentage of 32.

e 6 as a percentage of 60.

f 8 as a percentage of 12.

8 Calculate $28\ \text{cm}$ as a percentage of $3\ \text{m}$.

9 What percentage of 2 hours is 48 minutes?

Applications

10 A car weighs $1650\ \text{kg}$ and a truck weighs 3 tonnes. Express the mass of the car as a percentage of the mass of the truck.

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11 Find:

a $443.22\ \text{km}$ as a percentage of $890\ \text{km}$

b $164\ \text{cm}$ as a percentage of $8.2\ \text{m}$

c $\$6$ as a percentage of $\$375$

d $\$966$ as a percentage of $\$27\,600$

e $45.5\ \text{kg}$ as a percentage of $65\ \text{kg}$

f $8550\ \text{mL}$ as a percentage of $28.5\ \text{L}$

g $36\ \text{min}$ as a percentage of $4\ \text{hrs}$

h $108.9\ \text{mL}$ as a percentage of $330\ \text{mL}$

12 There are 11 boys and 14 girls in a class.

a Find the total number of students in the class.

b What percentage of the class are boys?

c What percentage of the class are girls?

13 A box of fruit contains 8 apples, 8 mandarins, and 4 pineapples.

a Find the total number of pieces of fruit inside the box.

b What percentage of the box are apples?

c What percentage of the box are mandarins?

d What percentage of the box are pineapples?

14 In a factory producing light bulbs, it is found that 27 out of a batch of 300 light bulbs are faulty. What percentage of the light bulbs are faulty?

15 Deborah scored 40 out 50 in her history test and 57 out of 75 in her mathematics test.

a What percentage score did Deborah get for her history test?

b What percentage score did Deborah get for her mathematics test?

c On which test did Deborah get a greater percentage score?

16 Justin and Georgia handed out fliers to advertise for their new concert.

Justin handed out 120 of his 400 fliers while Georgia handed out 360 of her 600 fliers.

a What percentage of his fliers did Justin hand out?

b What percentage of her fliers did Georgia hand out?

c Who handed out a higher percentage of their fliers?

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